

Research Idea: Arp_220

ALMA Thesis Planner (automated pipeline)

Sign first, depth second: mapping the 650 μm continuum and the CO(4–3) line-to-continuum ratio across Arp 220’s two nuclei at ~ 30 pc

Data used (all of it, and only it): the single Band 8 member OUS ‘uid://A001/X1289/X1e6’ of ALMA project 2017.1.00735.S (PI Rangwala; public since 2019-11-22), one pointing on Arp 220 at RA, Dec = 233.73843, +23.50322; spectral windows 451.9–453.8 / 453.6–455.6 / 463.8–465.8 / 465.6–467.6 GHz; tabulated angular resolution 0.086'' (≈ 30 pc at Arp 220’s distance, per the archive annotation), native velocity resolution 0.746 km/s, t_int 302 s, line sensitivity 8.336 mJy/beam @10 km/s, continuum sensitivity 0.2723 mJy/beam. QA2 image products and pipeline-calibrated measurement sets from the ALMA Science Archive. The CO J=6–5, CO J=3–2, ^{13}CO 4–3 and ^{13}CO 6–5 data named in the proposal abstract belong to other, unlisted member OUS and are **not** in this dataset; multi-line excitation modeling and the $^{12}\text{C}/^{13}\text{C}$ isotopic ratio are explicitly out of scope.

The line and its stated provenance: the target transition is ^{12}CO J=4–3, rest frequency 461.041 GHz, which the archive annotation places at ≈ 452.8 GHz *assuming* Arp 220’s systemic redshift $z = 0.018126$ ($cz = 5434$ km/s), putting it inside the 451.9–453.8 and 453.6–455.6 GHz windows. That line list was generated from general knowledge and must be verified against Splatalogue, with the sky-frequency placement re-checked against the actual MS header, before any imaging. The 463.8–465.8 and 465.6–467.6 GHz windows are annotated only as *tentative* candidates for high-J HC_3N (J=51–50, rest ≈ 464.0 GHz; J=52–51, rest ≈ 473.1 GHz), and the annotation is itself agnostic as to whether these would be redshifted or near-rest-frequency serendipitous detections — that identification is not established. Those windows are used here as additional continuum only; any HC_3N feature is reported as a tentative, clearly-flagged secondary result.

Headline deliverable: a spatially and spectrally resolved characterization of (a) the 650 μm continuum of the east and west nuclei — positions, deconvolved sizes, peak and integrated flux densities — and (b) the **line-to-continuum ratio** $R(v) = I_{\text{line}}(v)/I_{\text{cont,local}}$ of CO(4–3) across those nuclei, at the finest spatial scale the data actually support. Whether $R < 1$ (absorption) or $R > 1$ (emission), and where the sign changes, is a *measurement this thesis makes*, not an assumption it starts from.

Sign is not assumed. The proposal abstract’s “deep CO absorption” statement concerns CO 6–5 and 3–2 in other member OUS. That CO(4–3) also goes below the continuum here is an inference, and the plan branches on it explicitly rather than discovering the problem in week 9:

- **Absorption branch** ($R < 1$ over a nucleus or a set of positions): report apparent absorption depth $D(v) = -\ln R(v)$, its peak, the absorption-weighted velocity centroid relative to $cz = 5434$ km/s, the absorption FWHM, and PV cuts along each nucleus’s major axis.
- **Emission branch** ($R > 1$): report integrated line flux, moment-0/1/2 maps, and PV cuts

over the *same* positions, plus the continuum-referenced line-to-continuum contrast.

- **Mixed** (the likely and most interesting case): a sign map of ($I_{\text{line}} - I_{\text{cont}}$) per position and per velocity bin, with each branch's products computed on its own subset. The sign map is a headline figure in its own right — it locates where the dust brightness temperature exceeds the CO excitation temperature and where it does not.

Interpretation framing — brightness temperature, not front/back geometry. In these nuclei the 650 μm dust is optically thick and *co-spatial* with the CO, so this is not a foreground screen against a distant background source. A line appears in absorption here when the continuum brightness temperature exceeds the CO(4–3) excitation temperature along the same parcel of gas, and in emission when it does not. The discussion therefore leads with $T_{\text{B,cont}}$ versus T_{ex} and with the possibility that velocity structure in R reflects an excitation or temperature gradient within a rotating disk. The infall-versus-outflow question is named as the program's motivation and revisited only at the end, with the explicit statement that this geometry does not license the directional inference "redshifted absorption foreground gas falling in" — not even as a preferred reading.

Student work plan (~one semester):

1. (*Weeks 1–2*) *QA2 first.* Download the OUS and work from the **QA2 image products as the primary path**: continuum photometry, 2D Gaussian fits for the two nuclei, and a first look at the delivered cube. Restoring a 2017-cycle Band 8 pipeline calibration needs a matched CASA version and can silently eat two weeks; re-imaging with 'tclean' is an *enhancement* attempted only after the QA2-based measurements exist, and is abandoned on a written schedule if it stalls. Any self-calibration attempt is likewise bounded: a short ladder of solution intervals, and if none gives acceptable solution SNR at 302 s on source, self-cal is dropped and the failure documented with diagnostic plots.

2. (*Week 3*) *Go/no-go on the quantity that decides the project.* Measure the **per-beam continuum brightness map** — much of Arp 220's flux may be resolved out at 0.086'', so the per-beam peak, not the total flux, sets everything — and convert to $\sigma_{\text{R}} \approx \sigma_{\text{line}}/I_{\text{cont}}$ per 20 km/s bin. **Threshold, motivated:** the science needs to distinguish a shallow line-to-continuum feature ($|R - 1| \approx 0.25$, i.e. $D \approx 0.3$) from zero at $\geq 3\sigma$, and to tell a factor-of-two variation in depth across a nucleus from a flat one; both require $\sigma_D \approx 0.1$. Count beams meeting $\sigma_D < 0.1$ and declare a tier, in writing, in week 3:

- **Tier A** ($\gtrsim 20$ qualifying beams per nucleus): resolved maps of R, sign, centroid, and width.
- **Tier B** (a handful): spectra at named, tabulated positions — continuum peaks plus offsets along each disk's major and minor axes.
- **Tier C** (few or none): one spatially integrated spectrum per nucleus over the continuum-bright region, with beam-scale structure reported as upper limits. Tier C is a legitimate thesis outcome, written in now so marginal data are never over-resolved.

3. (*Weeks 4–6*) *Local continuum baseline, and the sign measurement.* The reference is taken **local** to the line, from line-free channels *within* the 451.9–453.8 / 453.6–455.6 pair, never extrapolated across the ~ 11 GHz gap. The dominant systematic is line-free-channel selection in a line-rich spectrum: quantify it by comparing at least two independent selections and propagating the difference. Then measure the sign of ($I_{\text{line}} - I_{\text{cont}}$) at every qualifying position and produce the sign map — this precedes and determines which products of step 5 get built. (No spectral-index measurement

is attempted: $\Delta\nu/\nu \approx 2.6\%$ across the window pairs predicts a dust-slope difference comparable to Band 8 spw-to-spw amplitude systematics, so the check is uninformative by construction and its days go here instead.)

4. (*Weeks 6–9*) *Cube, imaged narrow and binned from the start.* Image CO(4–3) **without continuum subtraction** — continuum subtraction would destroy the sign information this thesis is built on — restricted to the CO(4–3) velocity range and binned to $\sim 10\text{--}20$ km/s at the imaging stage. The native 0.746 km/s channels are unusable per beam at this sensitivity and integration time; full-resolution four-spw cubes are never produced. Continuum spws are imaged as continuum.

5. (*Weeks 9–11*) *Branch products.* Compute the absorption-branch and/or emission-branch quantities above at the tier selected in week 3. A **short table** (not a chapter) of CO column densities on an assumed T_{ex} grid, explicitly as lower limits, closes this section.

6. (*Weeks 11–13*) *Interpretation and write-up.* Describe how R, its sign, and its velocity centroid vary across each nucleus and how the two nuclei differ. Interpret in terms of T_{B,cont} versus T_{ex} and disk-internal excitation gradients first; treat infall/outflow as bounded discussion.

Limitations carried throughout: D is not an optical depth. CO(4–3) emission from the same co-spatial gas fills in any absorption along identical sightlines, and at 0.086'' this dataset has no emission-free reference position from which to calibrate that out — so every D is a lower bound contaminated by a spatially varying, unmeasurable term, on top of partial covering and saturation. The nuclei are optically thick well into the millimeter, so the "continuum" is a nearly opaque surface at unknown depth. The thesis writes *apparent absorption depth* or *line-to-continuum ratio* throughout, never τ . The **Band 8 absolute flux-scale uncertainty** ($\sim 10\text{--}20\%$) is carried explicitly in the continuum error budget: it affects the quoted nuclear flux densities, derived dust masses, and any brightness-temperature statement, and does *not* affect R or D, which are ratios taken within the same spectral setup — the thesis states which numbers it touches and which it does not.

How conclusions are phrased: the PI's own abstract states that ^{13}CO 6–5 is needed to *unambiguously* discriminate between absorption and outflow models, precisely because a single optically thick ^{12}CO line cannot separate genuine absorption from an emission/absorption blend nor break the optical-depth/covering-factor degeneracy. The thesis names the models it cannot exclude and identifies the specific ^{13}CO measurement that would close the question.

Deliverables: a 650 μm continuum image with nuclear photometry, deconvolved sizes, and an explicit flux-scale error budget; a documented QA2-vs-re-imaging and self-cal outcome; a per-beam σ_{D} map and the written tier decision; a CO(4–3) sign map; branch-appropriate line products at the selected tier; PV diagrams where supported; and a short lower-limit column-density table.

Why it is thesis-sized: one OUS, one pointing, one line, one measurement repeated over a bounded set of positions, with a tier rule that shrinks scope automatically and a sign branch that keeps the deliverables well-defined whichever way the line goes. QA2 products plus standard CASA and Python spectral fitting; no new observations, no ancillary datasets, no radiative-transfer code.